# Sound Calibration: Bench Report

Rice Auditory Benchmarking | 28 October 2026

**Purpose.** Make the measured sound components of 14 short tone pips more consistent across frequency. This is a bench test, not a validated animal stimulus.

**Setup and method.** An Audient iD14 played reconstructed 10 ms pips with 2 ms ramps at 96 kHz; an Earthworks M50 recorded the sound. Howard reported the microphone about 10 cm from the speaker, then moved it sideways twice by about 5 mm to improve high-frequency pickup. The final correction applies to that position and the same gain settings. A REED R8090 calibrator, reported set to 114 dB and fitted using an adapter, supplied a near-1 kHz reference.

The software reduces the digital level of stronger frequencies to bring their measured microphone components closer together; no frequency was boosted above the −36 dBFS base. The numbers below are *not* the digital playback settings.

**Results.** Two seven-second tests each presented 700 consecutive pips. The spread across the 14 frequency-group medians was **1.13 dB** and **0.77 dB**. These compare the recorded frequency component in the middle 6 ms, not every individual pip.

| Frequency (kHz) | Estimate (dB) | Frequency (kHz) | Estimate (dB) |
|---:|---:|---:|---:|
| 2 | 41.3 | 18.2 | 41.6 |
| 4 | 42.1 | 19.5 | 41.8 |
| 6 | 41.6 | 20.9 | 41.6 |
| 8 | 41.6 | 22.4 | 41.7 |
| 9 | 41.7 | 24 | 41.7 |
| 10.5 | 41.9 | 25.8 | 41.7 |
| 17 | 41.6 | 27 | 41.9 |

*Table:* Repeat-run frequency-group median estimates for the central component, expressed as equivalent-sine levels relative to 20 $\mu$Pa. Values assume the nominal 114 dB reference, unchanged linear gain and flat microphone response. Decimal places aid comparison; accuracy is not established.

**Limits and next steps.** These estimates are **not validated whole-pip dB SPL or standardized peak-equivalent SPL**. Background sound was substantial in full 10 ms windows; the microphone's individual electronic response, distortion and measurement uncertainty have not been verified. Equal measured pressure also does not establish equal perceived loudness. Recheck the reference, define the pip-level measurement, and repeat at the intended sheep-ear position before experimental use.

Evidence: Sound calibration page and measurement record.